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Integrating Knowledge Graphs with Symbolic AI: The Path to Interpretable Hybrid AI Systems in Medicine[☆]

Maria-Esther Vidal^{a,b,c,*}, Yashrajsinh Chudasama^{a,b}, Hao Huang^{a,b}, Disha Purohit^{a,b}, Maria Torrente^d

^a Leibniz University of Hannover, Hannover, 30159, Germany

^b TIB-Leibniz Information Centre for Science and Technology, Hannover, 30159, Germany

^c L3S Research Center, Leibniz University of Hannover, Hannover, 30167, Germany

^d Hospital Universitario Puerta de Hierro-Majadahonda, Madrid, 28222, Spain

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ABSTRACT

Knowledge Graphs (KGs) are graph-based structures that integrate heterogeneous data, capture domain knowledge, and enable explainable AI through symbolic reasoning. This position paper examines the challenges and research opportunities in integrating KGs with neuro-symbolic AI, highlighting their potential to enhance explainability, scalability, and context-aware reasoning in hybrid AI systems. Using a lung cancer use case, we illustrate how hybrid approaches address tasks such as link prediction—uncovering hidden relationships in medical data—and counterfactual reasoning—analyzing alternative scenarios to understand causal factors. The discussion is framed around TrustKG, which demonstrates how constraint validation, causal reasoning, and user-centric communication can support transparent and reliable decision-making. Additionally, we identify current limitations of KGs, including gaps in knowledge coverage, evolving data integration challenges, and the need for improved usability and impact assessment. These insights are not limited to healthcare but extend to other domains like energy, manufacturing, and mobility, showcasing the broad applicability of KGs. Finally, we propose research directions to unlock their full potential in building robust, transparent, and widely adopted real-world applications.

1. Introduction and motivation

The healthcare sector faces persistent challenges stemming from data silos, where vast amounts of heterogeneous data are stored in fragmented, structured, or semi-structured formats. This fragmentation hampers the seamless integration, analysis, and reuse of data, limiting the generation of actionable insights for decision-making [1,2]. These silos are fueled by diverse factors, including data volume and variety, restrictive access schemes, data sovereignty concerns, privacy and security constraints, as well as trust, legal, and ethical compliance issues (e.g., GDPR). Additionally, challenges such as missing data harmonization, low interoperability, and technical barriers further exacerbate the situation, hindering breakthroughs in knowledge discovery and the development of innovative healthcare technologies. As a result, data silos

restrict healthcare providers' ability to adopt data-driven approaches, potentially impacting the quality of care and life of patients [3,4].

Knowledge Graphs (KGs) have emerged as graph-based structures in AI, providing a structured framework for integrating heterogeneous data, capturing domain-specific knowledge, and enabling reasoning over complex relationships [5]. In healthcare, where decisions rely on synthesizing diverse information such as genomic markers, treatment histories, and patient demographics, KGs offer transformative opportunities to enhance AI systems. By encoding domain semantics, KGs support tasks requiring both predictive accuracy and interpretability, making them particularly valuable in medical contexts [1,2,6]. In medical research, hybrid AI approaches leveraging KGs have proven especially effective, facilitating the integration of heterogeneous clinical data and improving decision-making processes [3,4,7,8].

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* Corresponding author at: Leibniz University of Hannover, Hannover, 30159, Germany.

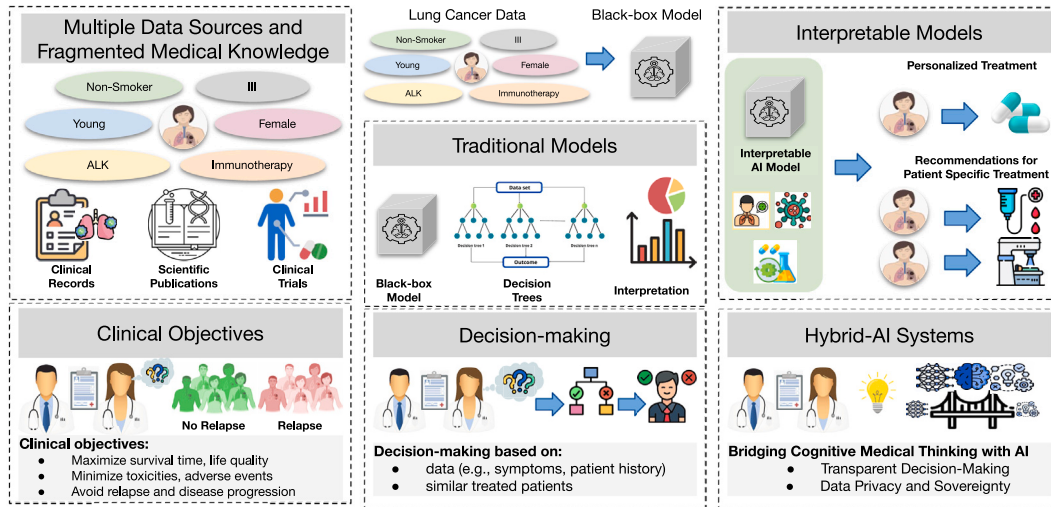
E-mail addresses: maria.vidal@tib.eu (M.-E. Vidal), yashrajsinh.chudasama@tib.eu (Y. Chudasama), hao.huang@tib.eu (H. Huang), disha.purohit@tib.eu (D. Purohit), m.torrente.prof@ufv.es (M. Torrente).

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(a) Scattered Data Sources and Fragmented Medical Knowledge hinders Clinical Objectives (b) Lack of transparency in AI models limits the effectiveness of data-driven decision-making systems. (c) Hybrid-AI Systems bridge reasoning with data-driven learning to provide patient-specific recommendations.

Fig. 1. AI models in medicine: This figure illustrates the transition from fragmented data and traditional black-box AI models to hybrid AI systems offering interpretable, personalized, and transparent decision-making. (a) Scattered data sources and fragmented knowledge hinder clinical objectives, such as minimizing relapse rates and improving patient outcomes. (b) Traditional black-box models lack transparency, limiting trust and effectiveness in decision-making processes and insights. (c) Hybrid AI systems integrate reasoning with data-driven learning to provide interpretable models, actionable treatment recommendations, and enhanced transparency while ensuring data privacy, scalability, and alignment with clinical and ethical requirements.

As illustrated in Fig. 1(a), clinical objectives in lung cancer (LC) management include maximizing survival and quality of life, minimizing toxic side effects, and preventing relapse or disease progression. However, traditional AI models (Fig. 1(b)) frequently function as opaque black boxes, lacking the transparency and interpretability essential for clinical decision-making. For example, treatment planning and relapse prevention require reliable AI systems capable of generating actionable and understandable insights. Hybrid AI systems integrating KGs address these challenges by combining symbolic reasoning and inductive learning to enhance both accuracy and interpretability (Fig. 1(c)). Some critical AI-based tasks include:

- **Relapse Risk Prediction:** Uncovering hidden associations between clinical factors and relapse risks to support personalized, preventive, and targeted interventions.
- **Personalized Treatment Optimization:** Using counterfactual reasoning to explore alternative treatment pathways, evaluate outcomes, and generate actionable, patient-specific recommendations.

Despite their potential, integrating KGs into AI pipelines poses significant challenges, including integrating heterogeneous data (*interoperability*), aligning evolving domain knowledge with AI models (*semantic integration*), enhancing performance of existing AI models (*reliability*), ensuring trustworthiness through comprehensible outputs (*interpretability*), and scaling systems to large healthcare datasets (*scalability*). Addressing these challenges requires innovative approaches to fully harness KGs' potential in advancing AI-driven solutions for real-world and complex applications.

This position paper highlights research challenges and insights drawn from practical experiences with hybrid AI systems in healthcare and other domains. It explores how KGs can enhance AI systems' interpretability and reliability while addressing inherent complexities. By focusing on lessons learned, this paper advocates for advancing the integration of symbolic reasoning and inductive learning to develop transparent and adaptable AI solutions tailored to user-centric needs. It identifies limitations of KGs, discusses their impact on hybrid AI systems, and proposes research directions to unlock their full potential in practical scenarios.

The paper is structured as follows: we first discuss research questions and key challenges in integrating KGs with AI (Section 2). Next, we present insights and lessons learned from applying KGs, including TrustKG, in healthcare scenarios (Section 3). Section 4 discusses the limitations of KGs, such as coverage gaps, evolving knowledge, usability, and tracing mechanisms—and proposes research directions to address these issues and enable the full adoption and effective utilization of KGs and hybrid AI systems.

2. Research questions and key research challenges for integrating KGs with hybrid AI

The integration of KGs into hybrid AI systems represents a transformative step in addressing the complexities of medical decision-making [1,2]. While this paper uses the medical domain as an exemplar case, the challenges outlined here are not unique to healthcare. Many other fields, such as energy [9,10], manufacturing [11–13], science [14], and mobility [15,16], face similar issues with fragmented, heterogeneous data and the need for transparent, knowledge-driven insights that support informed decision-making and foster collaboration across a wide range of stakeholders and end-users [17].

The medical domain is characterized by diverse and dynamic data sources, e.g., genomic markers, patient demographics, and imaging data, which are often fragmented across disconnected systems and heterogeneous vocabularies. This fragmentation creates challenges in interoperability, impeding knowledge-driven insights and limiting the scope for predictive modeling and personalized care. Addressing these issues is essential as the healthcare sector increasingly relies on AI models to generate insights to support a wide range of users, including patients, oncologists, radiologists, genomic researchers, and pharmacologists (Fig. 2).

In the context of lung cancer care, the need for hybrid AI systems that integrate symbolic reasoning with machine learning is particularly evident. These systems emulate the dual cognitive processes of human medical reasoning: rapid pattern recognition based on prior experience and deliberate, knowledge-based reasoning guided by formal medical protocols [7]. However, achieving such integration entails navigating several hurdles, including harmonizing diverse biomedical ontologies, ensuring semantic alignment, enabling efficient human-machine interaction, and ensuring transparency in AI-driven recommendations.

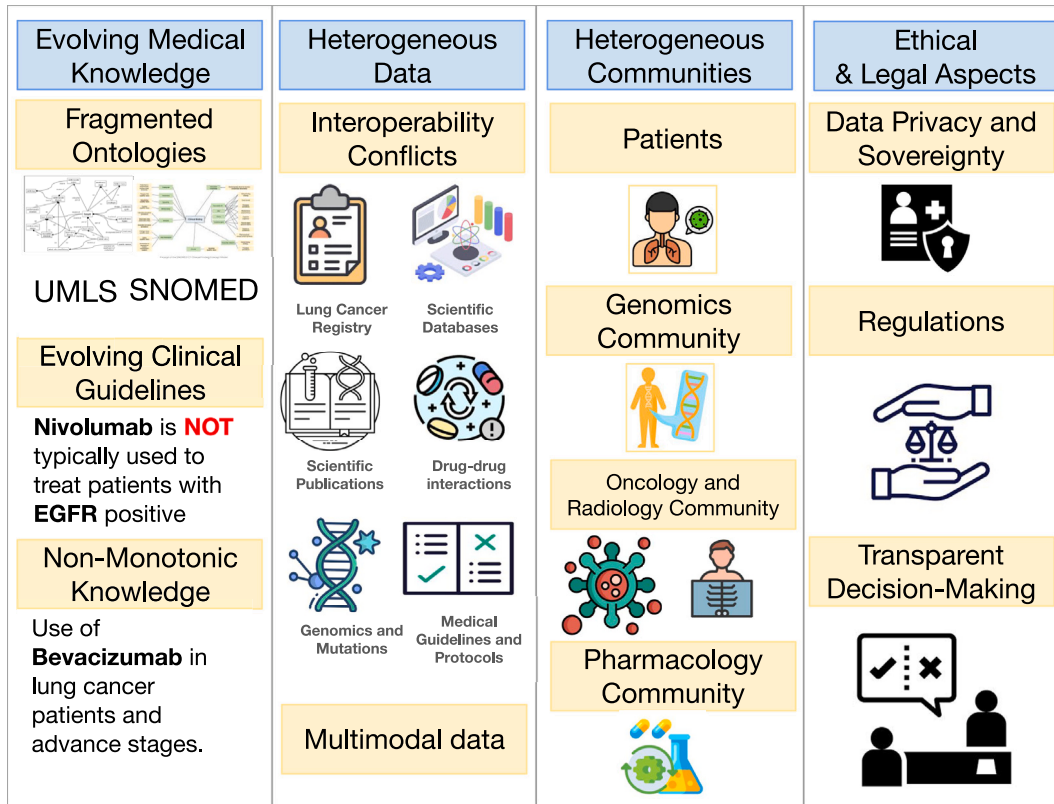


Fig. 2. Key medical challenges for hybrid AI systems. The figure summarizes four categories of challenges: (1) *Evolving Medical Knowledge*, highlighting fragmented ontologies, evolving clinical guidelines, and non-monotonic knowledge; (2) *Heterogeneous Data*, emphasizing multimodal data sources and interoperability conflicts; (3) *Heterogeneous Scientific Communities*, showcasing the diverse stakeholders and their varying perspectives; and (4) *Ethical and Legal Aspects*, addressing data privacy, sovereignty, regulations, and the need for transparent decision-making frameworks. Together, these challenges underscore the need for robust and interpretable hybrid AI systems in healthcare where domain knowledge is effectively modeled.

Addressing these challenges requires a multi-faceted approach to data integration, knowledge validation, predictive analytics, and actionable insight generation, tailored to diverse user needs. Moreover, these challenges highlight the domain-agnostic utility of KGs and their broader applicability across healthcare and beyond [1–3,10,16,18].

To explore the complexities of integrating KGs with neuro-symbolic AI for interpretable hybrid AI systems, we identify key research questions and challenges. These challenges arise from several factors. First, medical knowledge is evolving and often fragmented, requiring continuous updates and integration. Second, there is a pressing need to harmonize heterogeneous data sources to enable seamless interoperability, a challenge that extends across domains beyond medicine. Third, the diverse priorities and perspectives of multidisciplinary scientific communities complicate alignment and collaboration, an issue relevant to many fields. Finally, addressing critical ethical, legal, and regulatory considerations is essential for ensuring trustworthy and compliant systems (Fig. 2).

2.1. Research questions (RQs)

The integration of KGs into hybrid AI systems presents numerous unresolved research questions (RQs) that include data representation, reasoning capabilities, user interaction, and ethical considerations. These questions are particularly relevant in domains characterized by rapidly evolving knowledge, and heterogeneous and fragmented data.

- **RQ1: Evolving Knowledge:** How can hybrid AI systems manage evolving clinical guidelines and fragmented ontologies to ensure accurate and up-to-date recommendations for diagnosis and treatment planning? What methods can enable the effective representation and integration of non-monotonic knowledge, such

as changes in the recommended use of treatments (e.g., Bevacizumab for lung cancer), into KGs?

- **RQ2: Heterogeneous Data:** How can KGs effectively integrate multimodal data sources, such as clinical registries, scientific databases, drug–drug interactions, and genomics data while resolving interoperability conflicts? What data validation and harmonization techniques can ensure high-quality knowledge representation across diverse data formats in KGs?
- **RQ3: Heterogeneous Scientific Communities:** How can hybrid AI systems mediate the diverse perspectives and objectives of stakeholders such as genomics experts, oncology and radiology professionals, pharmacologists, and patients? What mechanisms can facilitate collaborative decision-making and ensure that KGs support the diverse needs of these user groups?
- **RQ4: Ethical and Legal Aspects:** How can hybrid AI systems ensure data privacy and sovereignty while complying with regulations such as GDPR, and enable meaningful insights? What transparent decision-making frameworks and KG-based approaches are needed to build trust and promote ethical AI?

2.2. Research challenges (RCs)

The research challenges (RCs) identified here stem directly from the research questions and represent critical barriers to effectively integrating KGs with hybrid AI systems. These challenges are foundational to advancing KG research and also provide essential context for the lessons learned, as illustrated through TrustKG in Section 3. TrustKG serves as the basis for discussing lessons learned, offering insights into practical approaches for addressing these challenges in KG-based systems and applications.

RC1: Data Integration and Interoperability. Data in medicine is inherently multimodal and scattered across various sources, such as patient registries, imaging databases, genomic datasets, and drug interaction records. These sources often rely on disparate schemas and vocabularies, leading to significant challenges in harmonizing and integrating the data into a unified framework [2,19]. For example, *Hypothyroidism* is defined as a syndrome in UMLS¹ or a disease in the Disease Ontology² and the Experimental Factor Ontology.³ Additionally, Wikidata⁴ and DBpedia⁵ provide definitions complementary to the ones provided by biomedical ontologies and vocabularies. Addressing these challenges requires collaboration among diverse stakeholders to define shared semantic representations and establish interoperability standards. Aligning terms across existing conceptual frameworks requires sophisticated mapping techniques and semantic reconciliation. KGs offer a promising solution by enabling the integration of heterogeneous data sources through shared semantic representations, facilitating seamless interoperability and data exploration [1]. This challenge aligns with research question RQ2, as it involves integrating data sources while ensuring interoperability and resolving schema conflicts.

RC2: Validation of Evolving and Process-Based Constraints. Medical guidelines and treatment protocols are frequently updated as new clinical evidence emerges, creating a dynamic landscape of evolving constraints [20]. Ensuring that these updates are validated against existing processes is critical to maintaining consistency and compliance. For instance, validating that a new drug recommendation does not conflict with previously established treatment plans requires the ability to model and enforce logical constraints dynamically. To address this, formal methods such as SHACL [21] and OWL [22] can be leveraged to define and validate evolving constraints. Hybrid AI systems can combine these symbolic techniques with machine learning to monitor, predict, and adapt to changes in clinical guidelines, ensuring their alignment with real-world applications [23]. This challenge directly aligns with research question RQ1, focusing on maintaining consistency and compliance as knowledge evolves and processes change.

RC3: Managing and Representing Evolving Medical Knowledge. The rapid pace of medical research generates a continuous stream of new knowledge, often resulting in fragmented and sometimes conflicting knowledge [2,24]. For example, disease definitions and treatment recommendations vary across biomedical ontologies. Representing and managing this evolving knowledge requires the creation of dynamic and adaptive KGs capable of accommodating updates and resolving contradictions [25]. Symbolic systems capable of handling non-monotonic changes while maintaining temporal consistency and computational feasibility [26], can maintain consistency in KGs, ensuring that users have access to up-to-date knowledge for decision making. In addition, techniques for detecting and resolving inconsistencies in medical knowledge need to be developed to improve reliability [27]. This challenge directly relates to research question RQ1, which focuses on managing evolving knowledge to provide accurate and up-to-date recommendations.

RC4: Modeling Positive, Negative, and Counterfactual Knowledge. In medicine, understanding not only what is true (positive knowledge) but also what is false (negative knowledge) and what could have been true under different circumstances (counterfactual knowledge) is essential [28]. For example, knowing that a particular drug is not effective for a subset of patients or understanding the potential outcomes of an alternative treatment pathway are critical for personalized medicine. Capturing this knowledge requires extending KGs to represent negations [23] and hypothetical scenarios [29], which can be

achieved by combining symbolic representations with probabilistic reasoning. Hybrid AI systems can also use counterfactual analysis to provide actionable insights, enhancing the interpretability and trust of medical recommendations [30]. This challenge relates to research question RQ1, as it involves representing and integrating evolving knowledge, including counterfactual and negative scenarios, to support personalized recommendations.

RC5: Ensuring Ethical and Legal Compliance. The sensitive nature of medical data necessitates strict adherence to privacy regulations, such as GDPR in the EU, HIPAA in the USA, and compliance with ethical guidelines. Challenges include enforcing data sovereignty, ensuring secure data sharing, and maintaining transparency in decision-making [31]. For example, AI models must trace and document how data was used to arrive at specific recommendations to ensure accountability [32]. Addressing this requires developing formalisms for encoding privacy rules and mechanisms for ensuring compliance, such as partitioning KGs into public and private sections. Transparent and interpretable AI systems must also be designed to foster trust among stakeholders, including patients and medical practitioners [33]. This challenge supports research question RQ4, emphasizing the need for frameworks that balance compliance with enabling actionable insights.

RC6: Providing Effective User-Centric Communication. Medical decision-making involves a wide range of users, from patients and clinicians to policymakers and researchers, each with unique communication needs [34]. For instance, a patient might require a simplified explanation of a treatment plan, whereas a clinician might need a detailed causal analysis. Bridging these needs demands the development of human-centric interfaces that enable intuitive exploration of insights derived from KGs [17]. Furthermore, symbolic reasoning can support the generation of explanations tailored to different user profiles. By integrating these capabilities into hybrid AI systems, it becomes possible to present actionable insights in ways that are comprehensible and valuable to diverse users and needs [34]. This challenge aligns with research question RQ3, focusing on supporting collaboration and addressing the varied perspectives of heterogeneous scientific communities.

RC7: Ensuring Scalability and High Performance. The scale of biomedical data is vast, encompassing heterogeneous registries of patients, drugs, and genetic markers [2,8]. Efficiently integrating, querying, and reasoning over this data in real-time poses significant technical challenges [4]. For example, executing complex queries across large-scale KGs demands high-performance algorithms and optimized storage solutions. Addressing these challenges requires leveraging distributed computing techniques, such as graph partitioning and parallel processing [35], alongside scalable ML models. Additionally, developing lightweight reasoning methods can ensure that hybrid AI systems maintain efficiency without compromising accuracy, enabling their deployment in high-stakes medical scenarios. This challenge sustains all research questions, as scalability and performance are essential for integrating heterogeneous data, managing evolving knowledge, and supporting diverse stakeholders effectively.

The lessons learned from addressing these challenges, as demonstrated through TrustKG in Section 3, provide practical insights into how hybrid AI systems can operationalize KG-based solutions across diverse domains.

3. Insights and lessons from TrustKG

The integration of KGs with neuro-symbolic AI has demonstrated significant potential in addressing the complexities of medical decision-making [3,34,36]. In this context, TrustKG is a hybrid AI framework that bridges symbolic reasoning and inductive learning to deliver interpretable and reliable solutions tailored to healthcare needs and user-centric recommendations.

¹ <https://uts.nlm.nih.gov/uts/umls/concept/C0020676>.

² <https://www.informatics.jax.org/disease/DOID:1459>.

³ https://www.ebi.ac.uk/ols/ontologies/efo/terms?iri=http%3A%2F%2Fwww.ebi.ac.uk%2Fefo%2Fefo_0004705.

⁴ <https://www.wikidata.org/wiki/Q16501>.

⁵ <https://dbpedia.org/page/Hypothyroidism>.

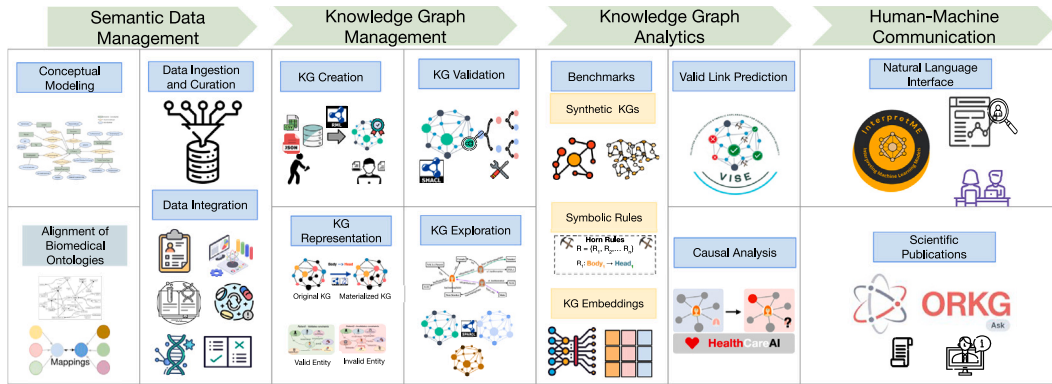


Fig. 3. TrustKG: A hybrid AI framework. *Semantic Data Management* integrates diverse biomedical data through modeling, ontology alignment, and semantic reconciliation. *Knowledge Graph Management* ensures consistency with tools for KG creation, validation, and exploration, enabling semantically rich and accurate representations. *Knowledge Graph Analytics* supports advanced tasks like link prediction and causal analysis by combining symbolic reasoning with synthetic KGs, fostering transparency and interpretability. *Human-Machine Communication* enhances accessibility with tools like InterpretME and ORKG Ask, ensuring actionable insights and fostering trust among diverse stakeholders. Together, these components demonstrate how KGs can operationalize hybrid AI systems to address complex, real-world challenges and provide user-centric, interpretable solutions.

TrustKG supports tasks such as valid link prediction, counterfactual reasoning, and human-centric communication, while addressing challenges related to data integration, evolving knowledge, and domain-specific requirements. TrustKG comprises interconnected layers that together support predictive modeling and decision-making (Fig. 3).

- ♦ **Semantic Data Management** harmonizes multimodal biomedical data through semantic alignment, ontology mapping, and named entity recognition and linking, ensuring consistency and interoperability across heterogeneous data sources.
- ♦ **Knowledge Graph Management** enables KG creation, validation, and exploration to maintain accurate and semantically rich knowledge representation and management at scale.
- ♦ **Knowledge Graph Analytics** combines symbolic reasoning and constraint validation with numerical learning for tasks like valid link prediction and counterfactual reasoning, thereby improving clinical decision-making, precision, and interpretability.
- ♦ **Human-Machine Communication** bridges technical outputs and user needs by translating results into natural language and linking insights to scientific literature through tools like InterpretME [34] and ORKG Ask [37], enhancing accessibility and usability.

TrustKG operationalizes research questions and challenges presented in Section 2 by addressing critical barriers through its layered architecture. The Semantic Data Management layer directly tackles RC1 (*Data Integration and Interoperability*) by harmonizing multimodal data sources, ensuring consistent semantic alignment and resolving schema conflicts. The Knowledge Graph Management layer contributes to RC2 (*Validation of Evolving Constraints*) and RC3 (*Managing Evolving Knowledge*) by enabling dynamic KG updates, integrity validation, and resolving contradictions in evolving medical knowledge. The Knowledge Graph Analytics layer addresses RC4 (*Modeling Counterfactual Knowledge*) by supporting tasks such as valid link prediction and counterfactual reasoning, thereby enabling personalized and interpretable recommendations. Furthermore, the Human-Machine Communication layer directly supports RC6 (*User-Centric Communication*) by providing tailored insights, linking results to scientific literature, and fostering trust among diverse users, including clinicians and researchers. Additionally, TrustKG integrates privacy-preserving mechanisms, contributing to RC5 (*Ethical and Legal Compliance*), and is designed for scalability, addressing RC7 (*Scalability and High Performance*) to ensure efficient reasoning over large-scale datasets.

TrustKG comprises two specialized systems, **VISE** [23] and **HealthCareAI** [30], both addressing critical challenges in implementing hybrid AI systems and offering innovative solutions to the research questions and challenges outlined in Section 2.

VISE addresses the challenge of plausible link prediction in KGs by identifying the most likely entity o' to complete triples of the form (s, p, o) while ensuring domain constraints are validated through SHACL shape schemas. It integrates symbolic learning, derived from SHACL validation and mined rules, with KG Embedding models, enhancing both prediction accuracy and interpretability. The hybrid AI methods implemented in **VISE** were evaluated on synthetic KGs generated from lung cancer clinical records, demonstrating its effectiveness in combining symbolic and numerical learning [23]. In this study, clinical guidelines for drug administration were represented as SHACL shapes, and link prediction tasks were solved using low-dimensional representations of the KGs learned by various KG embedding models, including (e.g., TransE [38], TransD [39], TransH [40], and RotatE [41]). The results highlighted how the representation of factual statements in KGs significantly affects model performance. Explicitly representing valid and invalid links in KGs yielded notable improvements in predictive accuracy. These findings corroborate that VISE effectively leverages the semantics of integrity constraints and mined rules to enhance the performance of KGE models, offering a more precise and interpretable solution to the valid link prediction problem in hybrid AI systems.

HealthCareAI addresses the challenge of counterfactual prediction by integrating symbolic reasoning, causal model learning, and counterfactual analysis. The approach enriches input KGs with domain knowledge derived from symbolic reasoning and metadata, combining statistical models and large language models (LLMs) to construct a causal graph that supports accurate and interpretable counterfactual predictions. This hybrid method enables the exploration of hypothetical treatments, providing actionable and context-aware recommendations. The effectiveness of **HealthCareAI** was evaluated on synthetic lung cancer (LC) KGs of varying sizes [30]. Ground truth counterfactuals for biomarkers were generated based on interventions on smoking type for each patient. The approach was compared against several baselines, including: (i) Statistical causal discovery methods (PC [42], FIC [42], GES [43]); (ii) LLM-based methods without considering domain knowledge; (iii) LLM-based methods incorporating domain knowledge; and (iv) Expert-designed causal graphs for causal model learning. Experimental results demonstrated the benefits of integrating semantics and domain-specific knowledge into causal learning pipelines. By leveraging metadata and mined rules (learned using symbolic learning), **HealthCareAI** achieved significant improvements in

the accuracy of causal models. The inclusion of semantic information provided a structured representation of causal relationships, aligning predictions with human reasoning and facilitating clearer, contextually grounded decision-making. These findings highlight the critical role of semantics in enhancing the precision of counterfactual predictions and underscore its importance in human-machine communication and transparency.

3.1. Practical insights

The implementation of TrustKG provides several insights into hybrid AI systems. First, the explicit definition of valid and invalid edges in **VISE** significantly enhances the predictive accuracy of KG Embedding (KGE) models [23]. By incorporating domain-specific integrity constraints and mined rules, **VISE** not only improves performance but also enhances interpretability in link prediction tasks. Second, **HealthCareAI** demonstrates the importance of integrating semantic metadata, mined rules, and LLMs with causal models. This integration improves predictive capabilities while aligning with human reasoning to support counterfactual predictions [30]. Lastly, TrustKG emphasizes the need for robust human-machine communication. By translating predictive insights into natural language and linking results to relevant publications, it ensures accessibility and fosters transparency and trust among diverse TrustKG users [34].

3.2. Lessons learned

TrustKG has yielded valuable lessons for hybrid AI and KGs.

- ◊ **Explicit domain constraints:** Leveraging domain-specific rules and constraints significantly enhances the reliability and interpretability of KG-based predictions. By incorporating integrity constraints represented through SHACL schemas and mined domain rules, **VISE** ensures that predictive models not only align with established medical guidelines but also produce outputs that are semantically meaningful and contextually accurate. This approach demonstrates the value of embedding domain knowledge directly into the KG reasoning process, enabling hybrid AI systems to address domain-specific challenges effectively while maintaining transparency and trustworthiness.
- ◊ **Causal reasoning integration:** The integration of symbolic reasoning, LLMs, and statistical models shows great promise for tackling complex tasks such as counterfactual prediction. **HealthCareAI** exemplifies this by leveraging semantic enrichment, mined rules, and causal graphs to provide accurate and interpretable counterfactual outcomes. This integration facilitates a structured representation of causal relationships that align with human reasoning, enabling more actionable and insightful predictions. Additionally, the results demonstrate that hybrid approaches incorporating semantics can improve the performance of AI-driven decision-making systems.
- ◊ **Interoperability and scalability:** Building scalable systems capable of integrating heterogeneous data sources while maintaining high performance is critical for operationalizing KG-based solutions. TrustKG highlights the importance of designing systems that can handle diverse, multimodal datasets, such as clinical records, imaging data, and genomic information, without sacrificing computational efficiency. Furthermore, ensuring interoperability through shared semantic representations and standardization across data sources lays the foundation for seamless integration into real-world workflows.
- ◊ **Human-centric interfaces:** TrustKG underscores the need for user-friendly interfaces that translate complex technical insights into actionable, comprehensible information. By incorporating natural language generation and linking predictive results to relevant scientific literature, TrustKG facilitates collaboration

and trust among diverse stakeholders, including clinicians, researchers, and patients. Such interfaces not only improve accessibility but also enhance the ability of users with varying expertise to derive value from KG-based hybrid AI systems.

These lessons underscore the importance of symbolic reasoning, semantic integration, and user-focused design in developing hybrid AI systems. Symbolic reasoning enhances interpretability by embedding domain knowledge into AI workflows, while semantic integration ensures consistent and meaningful representations across heterogeneous datasets. Lastly, prioritizing user-focused design enables seamless adoption and fosters collaboration across diverse user groups, making hybrid AI systems more effective and impactful in addressing real-world challenges.

4. Limitations of KGs in AI and research opportunities

While KGs have advanced hybrid AI systems [1–3,7,36,44], several limitations hinder their full potential and present opportunities for further research [17].

4.1. Key limitations of KGs

Despite the advancements demonstrated through TrustKG, inherent limitations of KG-based models constrain their effectiveness and broader adoption in complex applications. These limitations, identified during the development of TrustKG, underscore critical issues that must be addressed to fully realize the potential of KGs within hybrid AI systems.

- ◊ **Coverage Gaps.** Sparse and incomplete knowledge in KGs limits their effectiveness in predictive modeling and reasoning tasks, particularly in data-scarce domains.
- ◊ **Evolving Knowledge.** KG models struggle with handling dynamic knowledge, complicating updates, consistency maintenance, and practical implementation.
- ◊ **Limited User-Centric Tools.** The lack of tailored KG-based tools and intuitive visualizations significantly restricts accessibility, effective exploration, and meaningful utilization for a wide range of non-technical audiences.
- ◊ **No Provenance Tracing.** Existing frameworks lack mechanisms for tracing data sources, reasoning processes, predicting models, and KG-driven pipelines, which significantly undermines trust, transparency, and accountability in hybrid AI-driven insights.
- ◊ **Unmeasured Impact.** The lack of standardized methodologies to evaluate the usability, efficiency, and effectiveness of KGs in real-world scenarios limits the ability to assess their contributions to decision-making processes and hinders broader adoption of KGs in AI models.

4.2. Future directions in KG-driven hybrid AI

Building on these limitations, the following research directions offer pathways for advancement in KGs:

- ★ **Knowledge Management.** Develop methods to handle sparse and evolving knowledge within KGs, ensuring consistency and reliability in dynamic domains.
- ★ **User-Centric KG-based Systems.** Create tools and visualizations tailored to diverse user needs, fostering intuitive exploration and interaction with KGs.
- ★ **Provenance and Accountability.** Devise tracing mechanisms for data integration and curation, logical reasoning, and prediction processes to enhance transparency in KG-based hybrid AI.

- ★ **Scalability.** Develop advanced hybrid AI architectures that integrate symbolic reasoning with machine learning to efficiently manage KGs, ensuring real-time querying and reasoning across multimodal datasets while maintaining computational feasibility.
- ★ **Privacy and Interoperability.** Develop hybrid AI systems that integrate privacy-preserving techniques, such as federated learning and secure KG embeddings, into KGs. Simultaneously, establish universal standards for ontology alignment and cross-domain data integration to improve the interoperability of KGs, enabling their seamless use across diverse applications in hybrid AI.

By addressing the identified limitations and advancing the proposed research directions, KGs can enhance the capabilities of hybrid AI systems in solving complex, real-world problems. Developing solutions for sparse and evolving knowledge, scaling systems for large and multimodal datasets, and embedding robust provenance tracing will ensure that KGs remain reliable, adaptable, and transparent.

Future efforts should prioritize user-centric innovations, such as tailored visualization tools and intuitive interfaces, to improve accessibility for diverse stakeholders. Additionally, establishing standardized metrics for evaluating the impact and usability of KGs will help refine their deployment and foster trust in AI-driven decision-making. Enhanced interoperability and privacy-preserving techniques, such as federated learning, can further support seamless integration across domains and compliance with regulatory frameworks.

As demonstrated by TrustKG, these advancements can transform KGs into powerful enablers of explainable and scalable hybrid AI systems. By focusing on these concrete objectives, KGs can unlock their potential to drive meaningful innovation in diverse fields, including healthcare, manufacturing, and energy. We hope this position paper inspires the scientific community to address these open problems and develop effective methods and methodologies to realize the full potential of KG-driven hybrid AI.

CRedit authorship contribution statement

Maria-Esther Vidal: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Yashrajsinh Chudasama:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Hao Huang:** Writing – original draft, Methodology, Investigation, Conceptualization. **Disha Purohit:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Maria Torrente:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Maria-Esther Vidal reports financial support, administrative support, equipment, drugs, or supplies, statistical analysis, travel, and writing assistance were provided by German National Library of Science and Technology. Maria-Esther Vidal reports a relationship with German National Library of Science and Technology that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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